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| <br>SCG, URSC | Minutes of Software DRC Meeting<br>SCG Checkout Software Systems' DRC | Doc. No: SCG-FM-GEN-MM<br>Issue Date: 01-March-2017<br>Issue: A Rev: 00 |
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| <b>Meeting Chaired by</b>  | Ms. Muthumani Rajan                                                                                                          |
| <b>Date   Time   Venue</b> | 08-10-2021   01:30 PM-04:30 PM   Intranet Jitsi Room <SWDRC><br>26-10-2021   01:30 PM-04:30 PM   Intranet Jitsi Room <SWDRC> |
| <b>Agenda</b>              | Software Design Review Committee meeting for<br><i>SPAcecraft Status Display And Commanding Software (SPASDACS) 1.0</i>      |
| <b>Presented by</b>        | Mr. Janardhana Reddy G                                                                                                       |
| <b>Member secretary</b>    | Ms. Sheena Jose                                                                                                              |
| <b>Attendees</b>           | List enclosed                                                                                                                |
| <b>Enclosures</b>          | Nil                                                                                                                          |

### Agenda for the Meeting

To review the design aspects of the software package, *SPAcecraft Status Display And Commanding Software (SPASDACS) 1.0*.

Chairperson greeted the members. Ms. M. S. Bhuvaneswari, team leader of the software design team, gave an overview on the need for developing SPASDACS software. She also highlighted on the advantages of SPASDACS w.r.t PASDACS, where PASDACS is a similar software being used for checkout activities of GEO Payload from 2008 onwards.

Mr. Janardhana Reddy G, designer of the software presented in detail the requirements and the design details.

***Committee recommends designers to look into the highlighted points (Sl. No 6, 8 & 13) and provide the close out report.***

Review deliberations are as follows:

| No. | Deliberations/Actions                                                                                                                                                  | Actionee  | Status         |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------|
| 1.  | Committee appreciated the Software for the extensive features available for analysis of the behaviour of Spacecraft subsystems visually.                               | NA        | Information    |
| 2.  | To ensure the consistency in the Database parameters, Committee suggested to automate SPASDACS software interface with ACSS instead of taking the parameters manually. | Designers | To be complied |

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| No. | Deliberations/Actions                                                                                                                                                                                                       | Actionee  | Status         |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------|
| 3.  | For configuring the visual attributes of the Spacecraft elements, the possible values like messages of status parameters and limits of analog parameter to be shown to the User through auto-completion features.           | Designers | To be complied |
| 4.  | Software shall be more intelligent and User-friendly to handle Digital and Analog parameters. Presently User needs to know the type of the parameter and different values it can have, which may be avoided.                | Designers | To be complied |
| 5.  | Unauthorized user who knows the Javascript can edit the visual element features, but may not be allowed to save.<br>Committee suggested not to allow this.                                                                  | Designers | To be complied |
| 6.  | Even if there is syntax error during the creation of visual element, it is brought to final display. Such visual elements are not operational as intended by the user. This may cause wrong analysis, which has be avoided. | Designers | To be complied |
| 7.  | User needs to configure for each and every parameter on the required TM stream. It is suggested to handle all TM streams by default and provision to mention the required TM stream for the selected parameter.             | Designers | To be complied |
| 8.  | Presently the first user registration is taken as administrator.<br>This is not acceptable. Designers can plan and design for User account administration.                                                                  | Designers | To be complied |
| 9.  | Software can have a logging feature to log all the changes in the visual attributes of the Spacecraft elements displayed to the User.                                                                                       | Designers | To be complied |
| 10. | The configuration changes made to a Spacecraft element, which is currently being displayed on a client, should be reflected automatically on all the other client displays.                                                 | Designer  | To be complied |
| 11. | All the diagrams, files and associated information should be stored in a central repository and import provision to be provided for the Users to retrieve and configure across projects.                                    | Designer  | To be complied |

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| No. | Deliberations/Actions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Actionee | Status         |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------|
| 12. | Interface with ACSS to be specified in the SRS and design details to be put in the SDD.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Designer | To be complied |
| 13. | <p>Committee expressed apprehension on the following:</p> <ol style="list-style-type: none"> <li>1. Features provided to the Users to configure the visual elements through a programming language environment. Designers shall look into it and User friendly interface shall be provided.</li> <li>2. Software does not validate the parameter mapped with the visual element. User has to take care of mapping.</li> <li>3. Due to usage of various open source libraries in the software, there is a concern on the resources consumed by the software. Designers clarified that during the execution of the software, resources consumed are very less. However committee suggested that designers shall provide a report on the load testing of the software.</li> </ol> | Designer | To be complied |
| 14. | Committee wanted a demo on the software in RISAT-1A Checkout to assess the User responsibilities in configuring the software.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Designer | To be complied |

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| <b>Prepared by</b>              | Ms. Sheena Jose                                                                                                                                    |  |
| <b>Reviewed and Approved by</b> | Ms. Muthumani Rajan                                                                                                                                |  |
| <b>Minutes Issued Date</b>      | 03-11-2021                                                                                                                                         |                                                                                      |
| <b>Circulation List</b>         | <ul style="list-style-type: none"> <li>• Attendees,</li> <li>• Software DRC Committee Members,</li> <li>• DHs &amp; Group Director, SCG</li> </ul> |                                                                                      |

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### Attendee List

| No. | Members Present              | E-mail               | Availability<br>08-10-2021 | Availability<br>26-10-2021 |
|-----|------------------------------|----------------------|----------------------------|----------------------------|
| 1.  | Ms. Muthumani Rajan          | mani@usrc.gov.in     | Present                    | Present                    |
| 2.  | Mr. Vithal Metri             | vmetri@usrc.gov.in   | Absent                     | Present                    |
| 3.  | Ms. Bhuvaneswari M. S        | msbhuvan@usrc.gov.in | Present                    | Absent                     |
| 4.  | Ms. Smitha Anirudha Gokhale  | smitadeo@usrc.gov.in | Present                    | Absent                     |
| 5.  | Ms. Jagadevi Patil           | jagadevi@usrc.gov.in | Absent                     | Present                    |
| 6.  | Ms. Bharathi K. R            | bharathi@usrc.gov.in | Present                    | Present                    |
| 7.  | Mr. Subbaraya Sastry C. V. R | cvrs@usrc.gov.in     | Absent                     | Absent                     |
| 8.  | Ms. Lalitha k                | lalithak@usrc.gov.in | Absent                     | Absent                     |
| 9.  | Ms. Jeeja K. H               | jeeja@@usrc.gov.in   | Present                    | Absent                     |
| 10. | Mr. Ankti Agarwal            | ankit@usrc.gov.in    | Absent                     | Absent                     |
| 11. | Ms. Sheena Jose              | sheena@usrc.gov.in   | Present                    | Present                    |
| 12. | Mr. Janardhana Reddy G       | janag@usrc.gov.in    | Present                    | Present                    |
| 13. | Mr. Abbi Maniu Singh         | abimaniu@usrc.gov.in | Present                    | Absent                     |
| 14. | Mr. Bhosle Suraj Bharat      | surajbb@usrc.gov.in  | Present                    | Present                    |
| 15. | Mr. Devireddy Pramod Reddy   | pramodr@usrc.gov.in  | Absent                     | Present                    |